

Subtractive Dithering Technique for Delta-Sigma Modulator

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Abstract—This paper presents a novel subtractive dithering technique used in delta-sigma modulators which improves the spectral integrity of the modulators while minimizing the tradeoffs due to the use of dither. Dithering is commonly used in delta-sigma modulators to eliminate idle tones. The proposed technique adds dither at input of the quantizer and subtracts the dither in the analog domain at input of the modulator and in the digital domain at the output of modulator. The simulation results prove that when compared to the conventional dithering technique, this new subtractive dithering technique achieves the idle tone-free result but does not have the traditional drawbacks such as increased integrator output swing and in-band noise floor degradation.

Keywords—Dither; Subtractive Dithering; Sigma Delta ADC; Idle tone;

I. INTRODUCTION

Delta sigma modulators have been researched and used widely for the past several decades. Combined with oversampling and noise shaping, the delta sigma modulator can achieve a signal to noise ratio (SNR) higher than 100dB. It has been widely used in applications such as digital audio, sensor interfaces and communications [1] [2] [3].

Delta-sigma modulators (DSM) commonly use dithering techniques to remove idle channel tones. Lower order delta-sigma modulators such as 1st or 2nd-order modulators are particularly prone to idle tones if the input voltage is constant or at a low frequency [4]. Other unwanted tones can also be generated if the integrator nonlinearity is taken into account, which is the case in the real circuit implementation, since the OTA used in the integrator will always have a signal-dependent DC gain. The frequency of these tones depends on the input voltage level, thus when input voltage varies, the tone will move in the frequency spectrum. When the tone frequency falls in the signal bandwidth of modulator, the performance of the delta-sigma modulator will be corrupted [4]. By adding dither into delta-sigma modulator, the unwanted tones can be eliminated. The dither signal is usually inserted at the input of the quantizer which is the same location where quantization noise is generated, as shown in Fig. 1. Thus the inserted dither is also shaped by the loop filter of the delta sigma modulator.

There are several drawbacks when dither is used. First, although dither is shaped by the delta sigma modulator loop, its residual power in the signal band will lead to higher in-band noise. Second, the added dither will be processed by the loop filter and the output swing of the integrator will inevitably

increase. The increased output swing can lead to higher power consumption, and in some cases, a reduction in the usable input signal range. The situation may become worse for low-distortion topologies where the loop filter only carries the error signal [5]. In this case, the dither signal increases the overall error signal by 2X or more, causing the output swing of the integrators to become much larger.

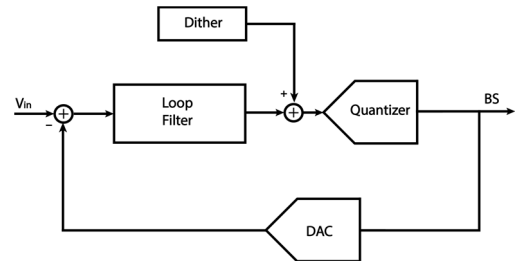


Fig.1 Delta sigma modulator with added dither.

To address above mentioned issues, this paper presents a subtractive dithering technique which subtracts dither in the analog domain at the input of modulator and also in the digital domain at output of the modulator. It allows the delta-sigma modulator to enjoy the benefits of dithering and avoid the drawbacks such as increased in-band noise or increased integrator output swing. The paper is organized as follows: Section II presents general dithering techniques used in delta-sigma modulators and Section III describes the proposed subtractive dithering techniques with detailed simulation results. The paper will be concluded in Section IV.

II. DITHERING IN DELTA-SIGMA MODULATOR

Fig. 2 shows the popular 2nd order CFF topology (cascade of integrators with feed-forward coefficients) modulator using the low distortion topology of [5]. The feed-forward branch feeds the signal into the input of the quantizer and linear analysis shows that the loop filter will only processes quantization noise [5]. Therefore, the output swing of integrators is dramatically reduced. This allows the modulator to operate at the lowest power possible. The delta sigma modulator shown in Fig. 2 is simulated in Matlab to explore the effectiveness of dithering. A multi-bit delta-sigma modulator with 8 levels of internal quantization is chosen to simulate the generalized case. To make the simulation more realistic, the nonlinearity of the integrator is also modeled in the Matlab simulation [6]. The DC gain of the OTA integrator is set at 80dB and the modulator is simulated with an AC input signal at -65dB with respect to full scale. Fig.

3 shows the FFT of the output of the modulator; the integrated in-band noise is found to be -111dB at an over sampling ratio (OSR) of 128. The mismatch of feedback DAC is not included in the simulation for simplicity. There is a clear evidence of unwanted tones appearing in the frequency spectrum. Fig. 4 shows a histogram of the output swing of both integrators.

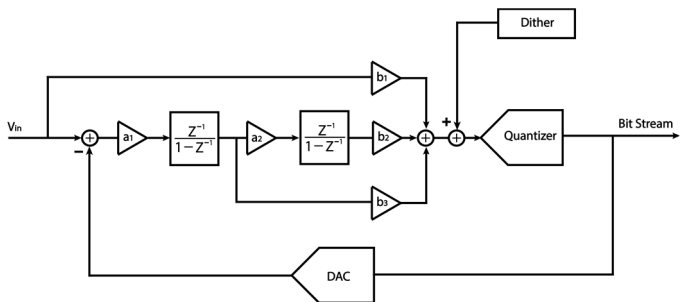


Fig.2 2nd order CIFF delta sigma modulator

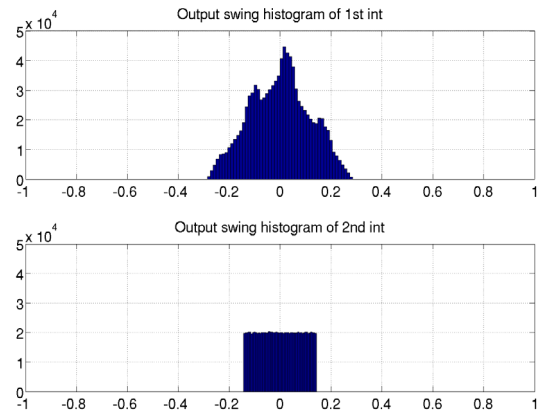


Fig.4 Histogram of output swing of two integrators

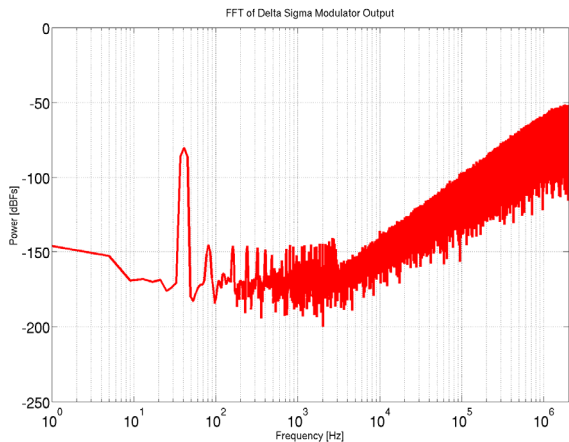


Fig.3 FFT of bit-stream of 2nd order CIFF DSM at -65dB input

The tonal problem mentioned above can be fixed by injecting dither into delta sigma modulator. Dither is usually added at input of quantizer, the same place where quantization error is generated. The dither signal randomizes the quantization error. Hence the quantization error becomes more like white noise. Dither and quantization error are shaped by loop filter; however, adding dither may increase the inband noise floor by several dB, depending on the RMS value of dither added. The dithered delta sigma modulator can be designed in such a way that the noise floor is dominated by thermal noise so the impact of dither can be neglected. In our simulation, an 8-level dither signal with a maximum value of ± 0.5 LSB and flat probability density function (PDF) is used. Fig. 5 shows histogram of the dither used in our simulation.

Fig. 6 shows the same delta sigma modulator as in fig. 3 but with dither added. It is clear that the tones are eliminated, which verifies the effectiveness of dithering. However, the integrated in-band noise has increased to -107dB which is 4dB higher due to the added dither. More importantly, the output swing of integrator increases dramatically as shown in Fig. 7. The output swings in Fig. 7 have almost doubled compared with Fig. 4, which puts a greater challenge on the integrator design.

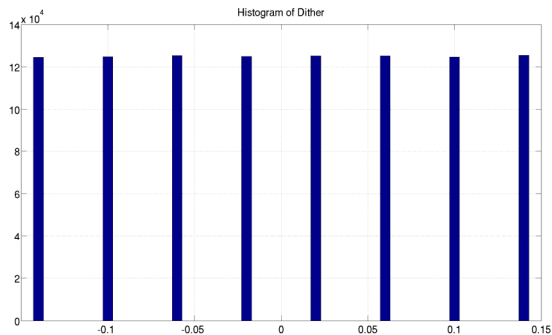


Fig.5 Histogram of dither used in simulation

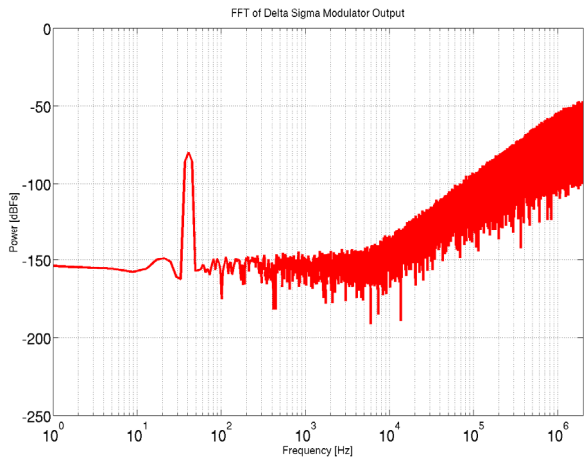


Fig.6 FFT of bit-stream of 2nd order CIFF DSM at -65dB input with dither

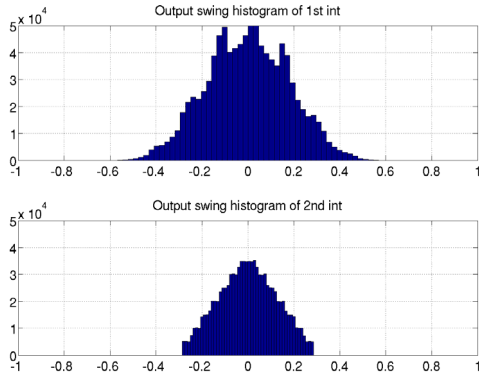


Fig.7 Histogram of output swing of two integrators with dithering

III. DITHER SUBTRACTION TECHNIQUES

The use of dither can eliminate unwanted tones and improve the performance of delta sigma modulator. However, added dither will introduce several problems. Beside the decrease of SNQR, the swing of the integrator is increased due to the added dither as shown in Fig. 7. The OTA used in the integrators needs to be designed to accommodate a higher swing requirement which leads to more power consumption. Alternatively, the loop filter coefficient a_1 and a_2 may be reduced and this will cause the in-band noise to increase even further.

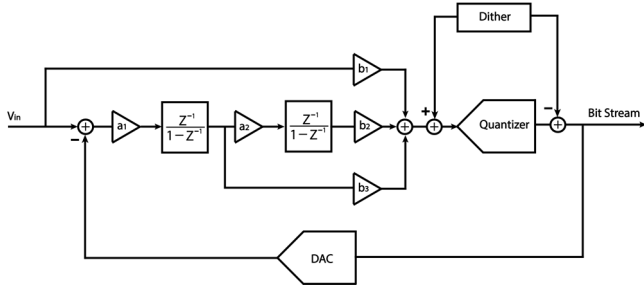


Fig. 8 Dither subtraction technique

The subtractive dither technique can be used to address the problem without compromising the benefits of dithering [7]. Fig.8 shows the idea of the subtractive dither technique. The dither can be added at input of quantizer and then subtracted afterwards. It is difficult to implement the subtraction right after quantizer, so the practical way to implement the subtraction is to move the dither subtraction to the modulator output, and then add another dither subtraction path to the input of the first integrator as shown in Fig. 9. Fig. 8 has exactly same mathematic model as Fig. 9. The subtraction of dither in the digital domain can be implemented easily and the subtraction at input of integrator can be implemented using a simple switched-capacitor DAC, assuming the modulator is a discrete-time delta sigma design. The FFT of the bit-stream of the modulator with dither subtraction enabled is shown in Fig. 10. It clearly shows that dither removes unwanted tones compared with Fig. 3. The simulated integrated noise is -111dB which is same compared with the non-dithered case. More importantly, the swing of integrators is almost unchanged compared with non-dithered case.

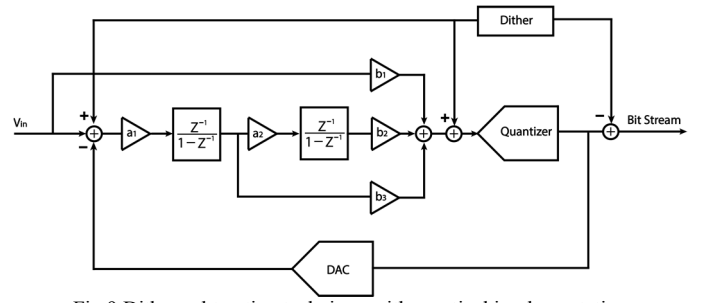


Fig.9 Dither subtraction technique with practical implementation

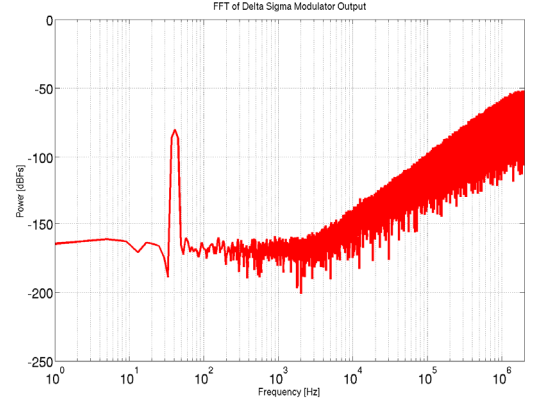


Fig.10 FFT of bit-stream of 2nd order CIFF DSM at -65dB input with dither subtraction

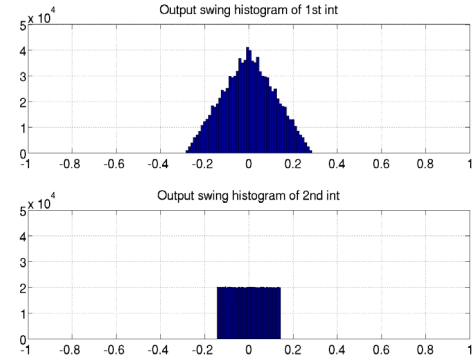


Fig.11 Histogram of output swing of two integrators with dithering subtraction

In conventional delta-sigma ADC design, single-bit quantization is a popular topology. However, the proposed subtractive dither technique cannot be applied directly to a 1-bit delta sigma modulator. The reason behind this is that when a single bit quantizer is used, the quantizer itself is highly nonlinear, so the added dither has a much higher possibility that it cannot change the output of the quantizer compared with a multi-bit quantizer. This will cause an error because the quantizer may be saturated by the dither addition, but the dither subtraction operation will still occur linearly at the modulator output and the first-integrator input. This may reduce the modulator performance by a substantial amount.

The above-mentioned problem can be fixed by introducing a signal dependent dither subtraction (SDDS) scheme which is shown in Fig. 12 [8]. One extra comparator is added to check the polarity of the original signal before adding dither; if the added

dither changes the comparator output, then the dither subtraction is activated; otherwise the dither subtraction is disabled. This signal-dependent dither subtraction technique ensures that dither will be subtracted only if it changes the output of the comparator.

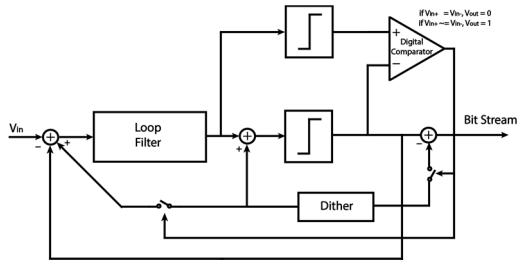


Fig.12 Signal dependent dither subtraction technique

Fig. 13 shows the FFT of the bit stream of a 2nd order CIFF modulator with a signal-bit quantizer. The AC input is set to -65dB with respect to full scale. The Matlab simulation includes the same integrator nonlinearity model as before. There is clearly evidence of tones in the baseband. Fig. 14 shows the FFT of the modulator using the conventional dithering scheme. The tones are removed but at the expense of higher noise floor and higher output swing of integrators. Fig. 15 shows the FFT of the modulator using the proposed signal-dependent dither subtraction scheme. It has tone free behavior and similar integrated noise level as the non-dithered version.

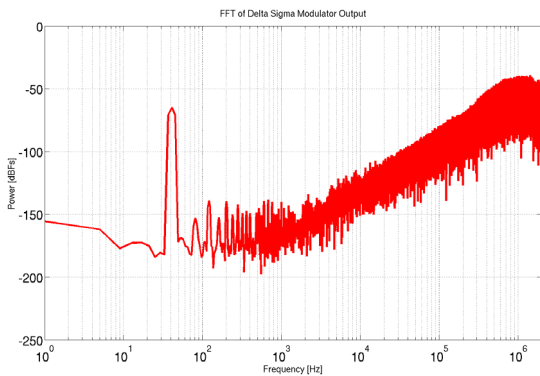


Fig.13 FFT of bit-stream of 2nd order single bit DSM

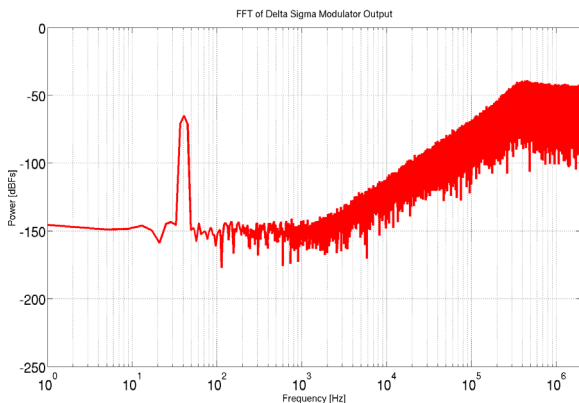


Fig.14 FFT of bit-stream of 2nd order single bit DSM with dither

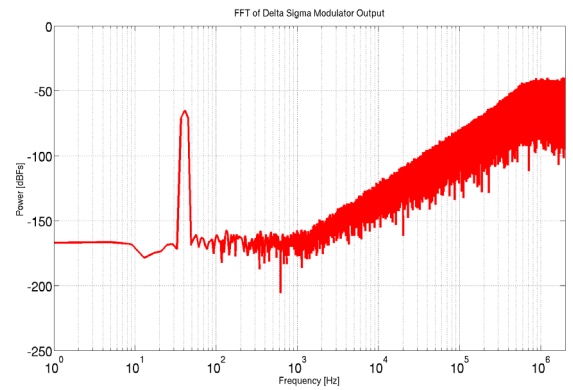


Fig.15 FFT of bit-stream of 2nd order single bit DSM with SDDS

IV. CONCLUSION

Dither is usually added into delta-sigma modulators to remove unwanted tones that result from quantization and internal non-linearities. However, the added dither also causes other problems, such as increased noise in the baseband and increased output swing of the integrators. The first issue decreases the performance of the delta sigma modulator and the later issue leads to higher power consumption and layout area in the OTA. The dither subtraction techniques proposed in this paper can address the above-mentioned problems. By subtracting dither at both the digital output of the modulator and also in the analog domain at the input of first integrator, the modulator is able to enjoy the benefits of dithering without the traditional drawbacks associated with dithering.

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